

$$\begin{array}{l}
 \text{Row 1} = \text{Row 1} - \frac{z_1}{z_1} \times \text{Row 3} \\
 \text{Row 2} = \text{Row 2} - \frac{(z_1 z_2 - z_2 z_1)}{(z_1 z_2 - z_2 z_1)} \times \text{Row 3}
 \end{array}
 \rightarrow
 \begin{bmatrix}
 1 & y_1/z_1 & 0 \\
 0 & 1 & (z_1 z_2 - z_2 z_1) \\
 0 & 0 & 1 \\
 0 & 0 & 0
 \end{bmatrix}$$

$$\begin{bmatrix}
 1 & y_1/z_1 & 0 \\
 0 & 1 & 0 \\
 0 & 0 & 1 \\
 0 & 0 & 0
 \end{bmatrix}
 \xrightarrow{\begin{array}{l} \text{Row 1} = \text{Row 1} - \frac{y_1}{z_1} \times \text{Row 2} \end{array}}
 \begin{bmatrix}
 1 & 0 & 0 \\
 0 & 1 & 0 \\
 0 & 0 & 1 \\
 0 & 0 & 0
 \end{bmatrix}$$

From prob 21, we know the final matrix is in reduced row echelon form. The 4th row has no pivot. Since A does not have a pivot in all rows, the columns of A do not span all of $\mathbb{R}^4$, i.e. a set of 3 vectors in $\mathbb{R}^4$ will not span all of $\mathbb{R}^4$. $\square$

For a set of n vectors in $\mathbb{R}^m$, we shall form a matrix A of size $m \times n$ comprising of the vectors. $\because m < n$, and by Theorem 1, we know the reduced row echelon form of a matrix is unique, we can conclude that in the reduced row echelon form of A , the last $(m-n)$ rows must be $\mathbf{0}$ zero rows. This is because:

- ① All non-zero rows must be above the zero rows.
- ② For each row, the leading entry must be in a column to the right of the columns of the leading entries of all rows above it.
- ③ For each column containing a leading entry, all positions below the leading entry position must contain 0.

From these conditions, it is easy to see that there can be at most n pivots for the first n rows from the top. Since every row of matrix A does not have a pivot, the set of n vectors cannot span $\mathbb{R}^m$ from Theorem 4. $\square$

Q3) Suppose A is a 4×3 matrix and b is a vector in $\mathbb{R}^4$ with the property that $Ax=b$ has a unique solution. What can you say about the reduced echelon form of A ? Justify your answer.

Solution: Let, a_1, a_2, a_3 be 4×1 vectors and let $A = [a_1 \ a_2 \ a_3]$. By Theorem 3, the equation $Ax=b$ has the same solution set as the system of linear equations corresponding to the augmented matrix $[a_1 \ a_2 \ a_3 \ b]$.